

LCARS TELEMETRY & RTC SYNC SYSTEM

1. EXECUTIVE SUMMARY & ARCHITECTURE OVERVIEW

The **Tufty 2040 LCARS System Telemetry & Clock Monitor** is a lightweight, dual-component application designed to stream real-time host metrics (CPU, RAM, Disk usage, Network IP addresses) to a Pimoroni Tufty 2040 display running MicroPython.

The system features an updated, resilient clock architecture that relies on the RP2040's hardware RTC (Real-Time Clock) initialized and drift-corrected periodically from the host via serial communication.

KEY SYSTEM HIGHLIGHTS & NEW FEATURES

- **One-Time / Periodic Time Synchronization:** Host transmits an initial timestamp at connection setup and sends recurring sync pulses every 60 seconds.
- **Automated RTC Drift Correction:** MicroPython firmware evaluates internal RTC offset and auto-corrects time drift whenever variance ≥ 1 second.
- **Timezone Offset Control:** Host sender accepts an explicit UTC offset parameter (`-o / --offset`) to seamlessly handle UTC+1, UTC+2, or negative offsets.
- **Standalone Autonomy:** The Tufty 2040 display maintains precise hardware time keeping even if telemetry data stream experiences temporary interruptions.
- **Full English Localization:** All logs, terminal outputs, error messages, and code comments are fully standardized in English.

2. COMMAND LINE USAGE & HOST SENDER (SENDER.PY)

The host script (`host/sender.py`) gathers telemetry using `psutil` and streams custom delimited payloads over the serial interface.

COMMAND SYNTAX

```
python host/sender.py <PORT> [--baud BAUD] [-o OFFSET] [-d]
```

ARGUMENT REFERENCE

ARGUMENT	TYPE	DEFAULT	DESCRIPTION
<code>port</code>	Positional	<i>Required</i>	Serial port identifier (e.g., <code>COM7</code> on Windows, <code>/dev/ttyACM0</code> on Linux/macOS).
<code>--baud</code>	Integer	<code>115200</code>	Baudrate for serial communication.
<code>-o</code> , <code>--offset</code>	Integer	<code>0</code>	Timezone offset in hours (e.g., <code>2</code> for UTC+2, <code>-5</code> for EST).
<code>-d</code> , <code>--debug</code>	Flag	<code>False</code>	Enables displaying incoming log messages sent back from Tufty 2040 over serial.

USAGE EXAMPLES

```
# Standard run for UTC+2 timezone on COM7
python host/sender.py COM7 -o 2
```

```
# Linux environment with debug logging and custom baudrate
python host/sender.py /dev/ttyACM0 --baud 115200 -o 1 --debug
```

3. PAYLOAD PROTOCOL SPECIFICATIONS

Communication uses single-line UTF-8 encoded text frames terminated with newline characters (`\n`). Key-value pairs are separated by colons (`:`) and delimited by pipes (`|`).

FRAME SPECIFICATIONS

FRAME TYPE	PAYLOAD EXAMPLE	FREQUENCY
TIME SYNC	INIT_TIME:1788278534	On startup + every 60 seconds
TELEMETRY	CPU:12.5 RAM_PCT:48.2 RAM_U:7.7 RAM_T:16.0 DISK_PCT:62.1 DISK_U:310.5 DISK_T:500.0 ETH:192.168.1.50 WIFI:N/A	Every 1 second
COMBINED	CPU:15.0 RAM_PCT:48.5 ... ETH:192.168.1.50 WIFI:N/A INIT_TIME:1788278534	Every 60 seconds (Merged frame)

4. FIRMWARE ARCHITECTURE & HARDWARE RTC MANAGEMENT (MAIN.PY)

The MicroPython script running on the Pimoroni Tufty 2040 utilizes non-blocking I/O (`select.poll()`) to ingest incoming serial streams while maintaining active button handling and screen rendering.

CLOCK SYNCHRONIZATION LOGIC

- Upon receiving a frame containing `INIT_TIME:<timestamp>` , the firmware converts the timestamp into UTC tuple representation using `time.localtime(ts)` .
- The current internal hardware RTC timestamp is calculated using `rtc.datetime()` and `time.mktime()` .
- If the absolute difference between the received timestamp and internal RTC time is ≥ 1 second, the RTC is updated via `rtc.datetime(...)` .
- If the drift is less than 1 second, the synchronization event is silently acknowledged, preventing jitter.

5. UI NAVIGATION & HARDWARE CONTROLS

BUTTON	SCREEN MODE / ACTION	DESCRIPTION
Button A	Telemetry Stats Screen	Displays real-time CPU, Memory, Disk gauges, and active Network IP addresses.
Button B	LCARS Clock View	Displays large formatted hardware RTC clock (HH:MM:SS), Date, and StarDate calculated directly from RP2040 RTC.
Button C	Toggle Light Mode	Switches between automatic lux sensor backlight control and manual mode.
Up / Down	Manual Backlight Adjust	Adjusts display brightness in 10% steps (switches backlight to Manual mode).